

Refutation of TLMC Conjecture 00000004396 (Sidorenko ladder-subdivision infinite families)

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1 The conjecture

The conjecture states that *all Möbius ladders are Sidorenko*, that *every odd subdivision of a tree is Sidorenko*, and that the union of these two families yields infinitely many new nontrivial Sidorenko graphs.

2 Definitions

Write $\text{hom}(G, H)$ for the number of graph homomorphisms $G \rightarrow H$.

Definition 1 (Sidorenko graph). *A finite graph G is Sidorenko if for every finite host graph H ,*

$$\text{hom}(G, H) |V(H)|^{|E(G)|} \geq \text{hom}(K_2, H)^{|E(G)|} |V(H)|^{|V(G)|},$$

equivalently $t(G, W) \geq t(K_2, W)^{|E(G)|}$ for every graphon W .

Definition 2 (Möbius ladder). *The Möbius ladder M_k has vertex set $\mathbb{Z}_{2k}$; vertex i is adjacent to $i \pm 1$ (rim edges) and to $i + k$ (spokes).*

3 The refutation

Theorem 1. *The Möbius ladder M_2 is not Sidorenko. Consequently the conjecture's first conjunct is false.*

Proof. M_2 has vertex set $\mathbb{Z}_4$ with rim edges $i \sim i \pm 1$ and spokes $i \sim i + 2$: every pair of distinct vertices is adjacent, so $M_2 \cong K_4$.

Take the host $H = K_2$. Since K_4 is non-bipartite, $\text{hom}(K_4, K_2) = 0$. But K_2 admits a homomorphism to itself (the identity), so $\text{hom}(K_2, K_2) \geq 1$ and the right-hand side $\text{hom}(K_2, K_2)^{|E|} 2^{|V|}$ is strictly positive. The Sidorenko inequality fails already at $H = K_2$, hence M_2 is not Sidorenko. $\square$

The same argument applies to every M_k with k even ($M_4, M_6, \dots$): when k is even the Möbius ladder is non-bipartite and therefore violates the $H = K_2$ instance of the bound (every Sidorenko graph must be bipartite). Odd- k ladders ($M_3 = K_{3,3}$, $M_5, \dots$) are bipartite and are not touched by this refutation, but the universal claim “*all* Möbius ladders are Sidorenko” is false.

4 Formalization

`lean4/Main.lean` defines `mobiusLadder n : SimpleGraph (ZMod (2*n+2))` (the classical ladder M_{n+1} ; the index shift keeps the modulus ≥ 2 so the graph is simple for all n without side conditions), the counting function `homCount` via `Nat.card`, the predicate `IsSidorenko`, and proves `mobiusLadder_one_eq_top` ($M_2 = K_4$ on $\mathbb{Z}_4$), `isEmpty_hom_K4_K2`, `isEmpty_hom_top_zmod4_K2` (transport of the previous emptiness along $\text{Fin } 4 \simeq \mathbb{Z}_4$), `homCount_mobiusLadder_one_K2` and the main theorem `not_sidorenko_mobiusLadder_one`. `Check.lean` audits axioms (`#print axioms`); only `propext`, `Classical.choice` and `Quot.sound` appear. Numerical verification is in `reproduce.py`.